Logistics Performance & Sales Analysis Dashboard

Dashboard Sections

The dashboard consists of four main sections, each addressing a specific analytical goal:

1. **Executive Financial & Profitability Overview** – Displays key KPIs such as total revenue, total profit, shipping cost, and profit margin to evaluate overall business performance.
2. **Operational Efficiency & Delivery Performance** – Focuses on shipment trends, delivery status (on-time vs delayed), and performance across shipment modes and regions.
3. **Geographical Flow & Carrier Analysis** – Analyzes shipment distribution across regions and evaluates carrier performance along with shipping costs.
4. **Customer Experience & Quality Analysis** – Examines customer satisfaction through ratings, delivery time, and service quality across regions and product categories.

Executive Financial & Profitability Overview

- Fact orders (Table)
- Order_ID
- Revenue_USD
- Profit_USD
- ShippingCost_USD
- Dim_Date (Date, Month, Year)
- dim_product (Product_Category)

Operational Efficiency & Delivery Performance

- fact_orders (Table)
- Order_ID
- Shipment_ID
- Delivery Status (calculated column)
- Dim_Date (Month, Year)
- dim_shipping (Shipment_Status, Shipping_Mode)
- dim_geography (Destination_Region)

Geographical Flow & Carrier Analysis

- fact_orders (Table)
- Order_ID
- ShippingCost_USD
- dim_geography (Origin_Region, Destination_Region)
- dim_shipping (Carrier_Type, Shipping_Mode)

- dim_product (Product_Category)

Customer Experience & Quality Analysis

- fact_orders (Table)
- Customer_ID
- Customer_Rating
- Order_Date
- Actual_Delivery_Date
- Delivery Days (calculated column)
- dim_product (Product_Category)
- dim_geography (Destination_Region)

Filters / Slicers

- Date (Day / Month / Year)
- Region / Country
- Product Category
- Shipment Mode
- Order Status

Visuals / Charts

Overview Page

- KPI Cards (Revenue, Profit, Orders, Margin)
- Line Chart (Sales Trend)
- Bar Chart (Region-wise Sales)
- Pie Chart (Order Status Distribution)

Logistics Page

- Bar Chart (Delivery Performance by Carrier)
- Line Chart (Delivery Time Trend)
- Stacked Column Chart (Order Status)

Sales Page

- Line Chart (Monthly Revenue)
- Clustered Bar Chart (Category Comparison)
- Tree Map (Product Contribution)

Interactivity

- **Cross-Filtering:** Selecting a region, product, or shipment automatically filters all other visuals in the report.

- **Tooltips:** Hovering over visuals displays additional details like revenue, profit, and order count for better insights.

Calculation & measures

1. Total Orders

Calculates the total number of orders recorded.

DAX:

```
Total Orders = COUNT(fact_orders[order_id])
```

2. Total Revenue

Calculates the total revenue generated.

DAX:

```
Total Revenue = SUM(fact_orders[revenue])
```

3. Total Profit

Calculates the overall profit.

DAX:

```
Total Profit = SUM(fact_orders[profit])
```

4. Profit Margin (%)

Shows profitability as a percentage of revenue.

DAX:

```
Profit Margin = DIVIDE([Total Profit], [Total Revenue], 0)
```

5. Total Quantity

Calculates total quantity of products sold.

DAX:

```
Total Quantity = SUM(fact_orders[quantity])
```

6. Average Order Value

Shows the average revenue per order.

DAX:

```
Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
```

7. Delivered Orders

Counts successfully delivered orders.

DAX:

```
Delivered Orders = CALCULATE(
COUNT(fact_orders[order_id]),
 fact_orders[order_status] = "Delivered")
```

8. Cancelled Orders

Counts cancelled orders.

DAX:

```
Cancelled Orders =
CALCULATE(
COUNT(fact_orders[order_id]),
fact_orders[order_status] = "Cancelled")
```

9. Pending Orders

Counts pending orders.

DAX:

```
Pending Orders =
CALCULATE(
 COUNT(fact_orders[order_id]),
fact_orders[order_status] = "Pending")
```

10. Delivery Rate (%)

Percentage of delivered orders.

DAX:

```
Delivery Rate = DIVIDE([Delivered Orders], [Total Orders], 0)
```

11. Average Delivery Time

Average number of days taken for delivery.

DAX:

Avg Delivery Time = AVERAGEX(fact_orders, DATEDIFF(fact_orders[order_date], fact_orders[delivery_date], DAY)

12. Late Deliveries

Counts orders delivered after expected date.

DAX:

Late Deliveries = CALCULATE(COUNT(fact_orders[order_id]), fact_orders[delivery_date] > fact_orders[expected_delivery_date])

13. On-Time Delivery Rate (%)

Percentage of orders delivered on time.

DAX:

On-Time Delivery Rate = DIVIDE([Total Orders] - [Late Deliveries],
[Total Orders],0)

14. Revenue per Customer

Average revenue generated per customer.

DAX: Revenue per Customer = DIVIDE([Total Revenue], DISTINCTCOUNT(fact_orders[customer_id]), 0)

15. Orders per Region

Counts orders by region.

DAX:

Orders per Region = COUNT(fact_orders[order_id])

16. Monthly Revenue

Revenue aggregated by month (use Date table).

DAX:

Monthly Revenue = [Total Revenue]

17. Year-to-Date (YTD) Revenue

Shows cumulative revenue for the year.

DAX:

YTD Revenue = TOTALYTD([Total Revenue],

dim_date[date])

18. Previous Month Revenue

Used for trend comparison.

DAX:

Previous Month Revenue = CALCULATE([Total Revenue], PREVIOUSMONTH(dim_date[date]))

19. Revenue Growth (%)

Shows growth compared to previous month.

DAX:

Revenue Growth % = DIVIDE([Total Revenue] - [Previous Month Revenue][Previous Month Revenue], 0)

20. Top Product by Revenue

Ranks products based on revenue.

DAX:

Product Rank = RANKX(ALL(dim_product[product_name]), [Total Revenue], DESC)

Export / Reporting Needs

- **Export Options:** PDF, Excel, Power BI Service shareable link
- **Scheduled Refresh:** Daily at 6:00 AM (Automatic data refresh from source)
- **Distribution:** Management team, operations team, and internal analytics team
- **Access Control:** Role-based access (Admin, Viewer)
- **Report Sharing:** Published via Power BI Service workspace
- **Data Source Update:** Connected to cleaned and transformed dataset via Power Query

Notes / Special Instructions

- Null and missing values are handled during data cleaning using Power Query
- Duplicate records removed to maintain data accuracy
- Data is normalized into fact and dimension tables (Star Schema)
- All visuals maintain consistent formatting and color theme across dashboards
- Date table is created and properly linked for time-based analysis
- DAX measures are optimized for performance and accuracy
- Filters and slicers are applied consistently across all report pages
- Proper relationships established between tables using primary and foreign keys
- Dashboard is designed for interactive analysis with drill-down and drill-through features
- Performance optimization applied (removed unused columns, optimized data model)